Boosting carbonylation chemistry through tunable active sites via d-block metal-p-block element synergy

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Carbonylation chemistry, involving the addition of CO to organic substrates, is crucial in the chemical industry, where molecular complexes of platinum group metals are widely applied as homogeneous catalysts.[1] However, the inherent challenges of separating homogeneous catalysts from their reaction mixtures after the catalytic reaction and their subsequent reuse highlight the need for the development of heterogeneous counterparts, which would enable a more facile application in the chemical industry. One strategy for synthesising high-performance solid-state catalysts involves the alloying of a metal with a non-metal, such as phosphorus.[2–4] Beside the reduction in noble metal content, the incorporation of a p-block element into a metal matrix, leads to the formation of highly defined and uniformly distributed active sites, which are often coordinatively unsaturated ("frustrated single-sites"), in analogy to homogenous molecular catalysts.[5] Herein, this so-called d-block metal-p-block element synergy is applied in the context of two highly important carbonylation reactions, namely the hydroformylation and alkoxycarbonylation.

In the context of hydroformylation catalysis, crystalline nanoparticles of metal rich $Rh_{17}S_{15}$ and metal lean Rh_2S_3 were prepared on a silica support and compared to the catalytic performance of purely Rh-based nanoparticles as reference catalyst in the hydroformylation of styrene. Hereby, a superior activity of the Rh_xS_y systems was observed in comparison to pure Rh, where the order of activity follows $Rh_{17}S_{15}$ > Rh_2S_3 > Rh. Through the combination of a plethora of analytic techniques, such as X-ray absorption and X-ray photoelectron spectroscopy (XAS and XPS), diffuse reflectance infrared Fourier transform spectroscopy (DRIFTS) using CO as probe molecule and electron microscopy techniques, together with local environment descriptors, unsupervised machine learning and density functional theory (DFT), we

could showcase the formation of spatially isolated $Rh^{\delta+}$ sites on an atomic level, which can easily be tuned with respect to their surface catalytic properties.[6] This concept was further investigated by synthesising supported Pd_xP_y nanoparticles, which were applied in alkoxycarbonylation reactions. These nanoparticles demonstrated superior activity compared to pure Pd and several commonly used homogeneous catalysts.

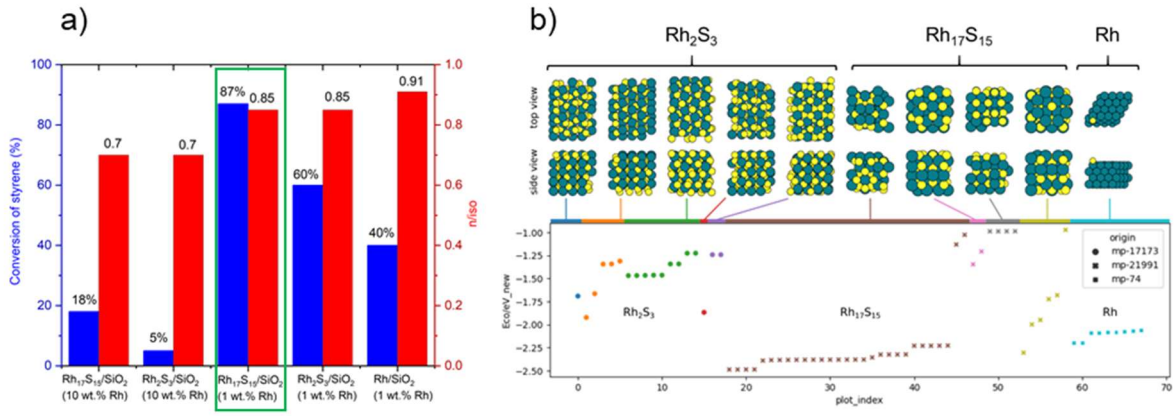

Figure 1: a) Conversion and selectivity for the hydroformylation of styrene using various Rh_xS_y and Rh catalysts (80°C, 40 bar ($CO:H_2$ 1:1) syngas pressure, 0.2 mol% catalyst concentration, 80 min) and b) active sites on Rh_xS_y surfaces with their corresponding CO binding energies ($E_{CO} = E^*_{CO} - E_{CO} - E^*$) considering the most favourable surface terminations.

To summarise, by introducing p-block elements into metal matrixes, highly active and isolated metal sites and structural motifs can be generated. The broad parameter space, involving metal and p-block element selection, phase compositions, and tailored synthesis methods presents a rich area for experimental investigation.

References

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